Service-Oriented Trust Modeling and Evaluation in O-RAN

Xiaoxue Deng, Yiheng Jiang, Yuwei Le, *Member, IEEE*, Jiaheng Wang, *Senior Member, IEEE*, Bin Sheng, *Member, IEEE*, Yongming Huang, *Fellow, IEEE*

***Abstract*—The open radio access network (O-RAN) incorporates network functions and applications from different vendors and domains to deliver always-online integrated services through multi-party coordination. Reliable O-RAN services rely on the joint effort from all parties who comply with the committed rules honestly. Therefore, trust evaluation on these O-RAN participants and their complex interactions has become critical to ensure trustworthy O-RAN services. Existing studies have explored trust in multiple domains, yet none of them can be used to characterize the trust in O-RAN through predefined rules and complex inter-party connections. To address these issues, we propose a service-oriented trust framework in O-RAN. We develop a trust model to establish a measurable representation of trust for heterogeneous O-RAN participants, and formulate trust relationship units (TRUs) to characterize diverse relationship patterns among them. We then quantify trust by integrating direct experience and verifiable knowledge, with contextual factors incorporated to adapt to varying service conditions. We further propose a TRU-based trust evaluation method to support trustworthy collaborative services. Finally, we implement an OpenAirInterface (OAI)-based prototype and validate the proposed framework through experiments. We show that the proposed framework can effectively distinguish unreliable service entities and improve O-RAN service reliability through trust-informed decisions.**

***Index Terms*—Open radio access network, trust model, trust evaluation, trust quantification.**

I. Introduction

As a foundational component of cellular systems, radio access networks (RANs) provide wireless access for user equipments (UEs) and connect them to diverse wireless resources. Future RANs are evolving towards open, virtualized, and intelligent architectures [1]. The open radio access network (O-RAN) serves as a representative trendsetter during this evolution [2]. Through standardized open interfaces and functional disaggregation, O-RAN supports the flexible deployment of network functions and applications, which are developed by different vendors and operated across different administrative domains. Unlike conventional RANs, where components from hardware to software are tightly integrated and supplied by a single vendor, O-RAN service delivery depends on coordination among heterogeneous entities with different functionalities, implementations, and administrative ownership, a process referred to as service integration [3]. In this context, the successful provision of RAN services relies on each involved entity correctly fulfilling its assigned role. Unreliable behavior by one entity (e.g., providing inaccurate information, failing to execute allocated functions, or exhibiting degraded performance) can lead to inappropriate control decisions, which in turn cause service disruptions and ultimately degrade user experience [4]. Therefore, trust has emerged as a fundamental imperative in modern information and communication technology (ICT) systems, particularly for next-generation mobile networks. The international telecommunication union (ITU-T) emphasizes in its standard Y.3052 [5] that trust is a critical factor in application and service transactions. International mobile telecommunications-2030 (IMT-2030) explicitly lists trust as a core requirement for future mobile communication networks [6]. The national institute of standards and technology (NIST) incorporates continuous trust evaluation as an essential principle of its zero trust architecture in special publication 800-207 [7], underscoring its critical role in security in future networks. Following this direction, the O-RAN alliance security work group has investigated the applicability of zero trust architecture to O-RAN and defined security requirements for O-RAN architecture elements, interfaces, and assets [8]. The O-RAN alliance next generation research group also identifies trustworthiness as an important consideration for cross-domain collaboration [9]. These developments indicate that trust has become an explicit consideration for O-RAN.

In recent years, a growing number of studies on trust have emerged across various disciplines, including sociology [10], psychology [11], and economics [12]. Trust, as defined in these studies, is a general belief held by an entity (i.e., trust subject) that another entity (i.e., trust object) will meet its expectations [13]. In the context of networking, trust can be interpreted from a more technical perspective. As defined in ITU-T Y.3052 [5], trust is viewed as the expectation that a cyber object will accomplish a given task in an expected manner to fulfill its intended purpose. Similarly, the authors of [14] define trust as the willingness to accept risk in an interaction. As noted in [15], trust in the network context extends beyond conventional security to encompass aspects like privacy, reliability, and resilience. This multi-faceted nature renders trust a highly complex construct. These definitions offer abstract conceptualizations of trust. However, it remains unclear what exactly trust is, nor whether a specific O-RAN service entity should be relied upon in a given service provision. Hence, to enable the establishment of trustworthy service collaborations in O-RAN,

X. Deng, Y. Jiang, J. Wang, B. Sheng, and Y. Huang are with the National Mobile Communications Research Laboratory, Southeast University, Nanjing 210096, China, and are also with the Purple Mountain Laboratories, Nanjing 210023, China (e-mail: {xdeng, cyluo, jhwang, sbdtt, huangym}@seu.edu.cn). Y. Le is with the Purple Mountain Laboratories, Nanjing 210023, China (e-mail: yueyuwei@pmlabs.com.cn).
Corresponding author: Jiaheng Wang.

trust must be measured and evaluated explicitly. As O-RAN involves heterogeneous participating entities and diverse forms of collaboration, this raises the fundamental question of how to systematically model and evaluate diverse trust relationships among them.

A. Related Work

To date, various studies have investigated trust in wireless networks. Ling *et al.* [16] defined trust as the probability that the trust object produces the desired outcome and fulfills its intended goal in a specific context. Liu *et al.* [17] proposed a trust model based on Bayesian theorem, modeling trust as random variables with specific probability distributions. Luo *et al.* [18] measured trust uncertainty through entropy, where a lower entropy value indicates a more certain trust state. He *et al.* [19] introduced a dynamic trust model for heterogeneous wireless networks, which described trust as the degree to which the observed behavior pattern aligns with a specific category. Xu *et al.* [20] proposed a trust evaluation method based on uncertainty theory, which utilizes several constraints to construct a linear programming problem for each trust path, and evaluates trust by solving these problems. Wang *et al.* [21] divided trust into interaction, energy, and recommendation components. The final trust value is computed as a weighted sum of these three components. Desai *et al.* [22] introduced a trust evaluation mechanism for subjects and objects associated along multiple paths, where each intermediate node propagates its computed trust value back to the previous hop. Trust values along the path are aggregated sequentially to obtain the overall trust from the source node to the destination node. Huo et al. [23] developed a graph neural network-based method for learnable trust propagation and composition. With respect to trust management in cross-domain service collaboration, Jorquera Valero *et al.* [24] proposed a reputation-based trust-as-a-service framework for the fifth-generation network in distributed multi-stakeholder environments, where accumulated reputation is used to identify reliable network operators. They further developed a trust management framework, in which trust scores are quantified according to providers' historical behavior and service level agreement (SLA) violations [25]. These works demonstrate that service fulfillment history can be converted into trust information and used to support collaboration across administrative domains.

In O-RAN, a growing but still limited body of academic work has explicitly examined trust or trustworthiness. Porambage *et al.* [26] reviewed security, privacy, and trust issues arising in O-RAN toward the six-generation network, and summarized the corresponding research trends and open challenges. Parra-Ullauri *et al.* [27] proposed an approach for managing the trustworthiness of artificial intelligence (AI) models throughout their lifecycle and demonstrated the approach through an AI-enabled xApp. Javed *et al.* [28] developed a blockchain-based reputation mechanism for federated learning in O-RAN, in which the accuracy, integrity, and quality of model updates are used to evaluate participating clients and support participant selection.

Despite the above progress, several limitations remain. 1) Most existing approaches target specific entities with scenario-specific evidence. ~~For instance, in [28], trust is evaluated based on the accuracy and integrity of participants' model updates, while in [29], trust is built upon five attack-related factors.~~ This makes evaluation results from different methods difficult to compare, propagate, or combine across O-RAN service relationships. 2) Existing works often provide limited support for representing diverse trust relationships arising in O-RAN service integration, as they typically focus on pairwise interactions or infer indirect trust through path-based propagation. Consequently, they may struggle to provide adequate trust inference in some scenarios. 3) Existing studies may face challenges in evaluating complex trust relationships. ~~Optimization-based or iterative approaches [20] often involve considerable computational processes, while learning-based methods [30] may provide limited interpretability regarding how trust results are derived, making trust decisions difficult to explain~~.

B. Contributions

To address the above issues, we propose a service-oriented trust framework in O-RAN. It enables modeling, quantification, and evaluation of diverse trust relationships among participants in collaborative service provision. Through this, we facilitate trustworthy and efficient service integration for O-RAN. The main contributions of this paper are summarized as follows.

- We design a trust framework for O-RAN that coordinates service-oriented trust modeling, quantification, and evaluation through a unified workflow, incorporating O-RAN characteristics.
- We establish a trust model in O-RAN, which provides a quantifiable basis for trust evaluation across vendors and domains. We set up multiple trust relationship units (TRUs) to represent basic trust relationships, with their flexible composition supporting the characterization of diverse trust relationships in O-RAN.
- We propose a context-aware trust quantification method, which combines experience-based service fulfillment evidence with knowledge-based information. We incorporate service context to implement reliable trust quantification under varying O-RAN operating conditions.
- We develop a service-oriented trust evaluation method, which evaluates diverse trust relationships among service entities ~~through a systematic process~~. We leverage such evaluation to support trustworthy collaborative service establishment and mitigate negative impacts from unreliable entities.
- We validate the proposed framework through a lab-built prototype. Experimental results demonstrate the effectiveness of the proposed framework in distinguishing unreliable entities, supporting service decisions, and improving O-RAN service performance.

The remainder of this paper is organized as follows. Section II presents the overall design and workflow of the proposed trust framework for O-RAN. Section III establishes the trust model. Section IV proposes the context-aware trust quantification

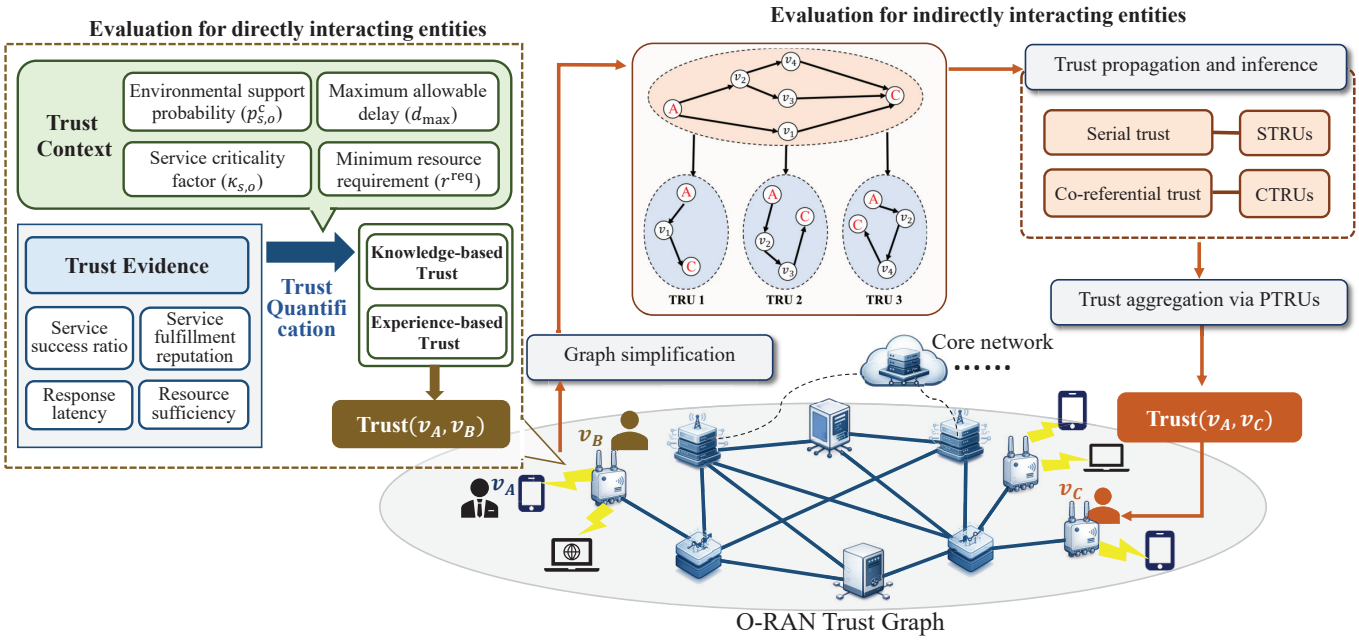


Fig. 1. Workflow of the proposed framework for service-oriented trust evaluation in O-RAN, where v_A represents the trust subject, while v_B and v_C denote the trust objects in direct and indirect evaluation scenarios, respectively.

method. Section V provides the service-oriented trust evaluation method in O-RAN. Section VI describes the prototype implementation and presents the experimental results. Finally, Section VII concludes the paper.

II. O-RAN Trust Framework

O-RAN service provision relies on the coordination of multiple heterogeneous entities from different vendors and domains, such as O-RAN central units (O-CUs), O-RAN distributed units (O-DUs), and service application instances. However, these entities may exhibit unreliable or malicious behaviors (e.g., tampering with service outputs, providing incorrect information), which can lead to service performance degradation or even service failures. Trust evaluation on these participating entities before establishing service collaboration serves as an effective approach to mitigate such risks by enabling reliable service decisions [25]. In this context, a service-oriented directed trust relationship is established from the evaluating entity, referred to as the trust subject, to the evaluated entity, referred to as the trust object. The associated trust value represents the degree to which an O-RAN entity can be relied upon by another entity to fulfill an assigned service role.

The proposed O-RAN trust framework consists of three components: trust model, trust quantification, and trust evaluation. The trust model provides the structural foundation for representing service-oriented trust relationships among O-RAN entities, and characterizes diverse relationship patterns through TRUs. The trust quantification component transforms service-related evidence under corresponding operating contexts into experience-based trust and knowledge-based trust, which together provide complementary views of the trust object. Based on these representations and quantified trust values, the trust evaluation component derives the final trust assessment for the trust object by considering different relationship structures among entities, which serves as the basis for trustworthy collaborative services.

The workflow of the proposed framework for service-oriented trust evaluation is as follows.

1) When an O-RAN entity intends to establish a service relationship with another entity that provides required service capabilities, it initiates trust evaluation to determine whether the target entity can be relied upon to fulfill the assigned service. For example, as illustrated in Fig. 1, a UE, denoted by v_A, may need to select an O-DU to provide radio access-related service. These candidate O-DUs may be deployed by different vendors and exhibit different service fulfillment capabilities. Among them, a candidate O-DU, denoted by v_B, may have existing service interactions with the UE. In contrast, another candidate O-DU, denoted by v_C, may be newly deployed and thus have no prior service relationship with the UE. Therefore, before establishing the service relationship, the UE performs trust evaluation on candidate O-DUs to determine whether they can provide reliable service as expected and selects an appropriate service entity based on the evaluation results. In this evaluation process, the UE v_A acts as the trust subject, while the candidate O-DUs v_B and v_C serve as the trust objects to be evaluated.
2) For the evaluation of v_B with prior direct interaction records, v_A performs direct trust evaluation based on the observed service-related evidence from historical interactions and the associated service context. Specifically, the collected evidence is quantified into experience-based and knowledge-based trust values, which are further integrated with the previous direct trust value of

v_A towards v_B to obtain the final trust evaluation result, denoted by Trust(v_A, v_B).

3) For the evaluation of v_C without prior direct interaction records, v_A performs indirect trust evaluation by exploiting trust relationships provided by other entities.
   a) v_A obtains relevant trust relationships from reachable entities within a predefined search range and constructs an evaluation-specific trust subgraph.
   b) v_A performs graph simplification on the constructed trust graph by removing redundant and overlapping trust relationships, and then decomposes the resulting subgraph into multiple TRUs.
   c) v_A propagates, infers, and aggregates available direct trust values through the corresponding TRUs to obtain the final trust evaluation result, denoted by Trust(v_A, v_C).
4) Based on the resulting direct or indirect trust value, v_A determines whether to establish or maintain the service relationship with the evaluated entity.

Through this workflow, the proposed framework enables systematic modeling, quantification, and evaluation of trust relationships among entities in participating collaborative service provision, thereby facilitating trustworthy service integration for O-RAN.

III. Trust Model

A. Trust Graph

The service dependencies and interactions among O-RAN entities from different vendors and domains give rise to diverse trust relationships. To provide a structured representation of these relationships, we formalize a directed trust graph, denoted as

$$G = (V, E, \mathbf{A}^{K|C}), \tag{1}$$

where V and E represent the sets of trust entities and direct trust relationships, respectively. The entity set $V = \{v_1, v_2, \ldots, v_N\}$, where $N (N \geq 2)$ is the total number of trust entities involved in the service delivery process in the O-RAN. These entities encompass all O-RAN entities involved in service provision, including UEs and network function instances deployed by different vendors or administrative domains, e.g., O-CUs, O-DUs. For a given trust relationship, we denote the entity initiating the trust relationship, i.e., the trust subject, as v_s and the entity being trusted, i.e., the trust object, as v_o. Each directed edge in the trust graph represents a trust relationship between adjacent entities. Formally, $E \subseteq \{e_{s,o} | v_s, v_o \in V, s \neq o\}$, where a directed edge $e_{s,o} \in E$ represents the direct trust relationship from v_s to v_o, established when v_s relies on a service, output, or function provided by v_o.

Additionally, $\mathbf{A}^{K|C} \in \mathbb{R}^{N \times N \times 2}$ is a tensor of trust assertions, where $\mathbf{A}^{K|C}_{s,o} \in \mathbb{R}^2$ is an element of this tensor, representing the trust assertion that characterizes the direct trust relationship $e_{s,o} \in E$. This assertion serves as the composite edge attribute of $e_{s,o}$. Mathematically, $\mathbf{A}^{K|C}_{s,o}$ is defined as

$$\mathbf{A}^{K|C}_{s,o} \triangleq \begin{cases} (T^{\text{exp}}_{s,o}, T^{\text{know}}_{s,o}), & \text{if } e_{s,o} \in E, \\ (0, 0), & \text{otherwise.} \end{cases} \tag{2}$$

Table I. Essential variables in the modeling and analysis.

Symbols	Explanations
G	Trust graph in O-RAN
V	Set of trust entities
E	Set of direct trust relationships
K, C	Trust evidence space and context space
$\mathbf{A}^{K\|C}$	Tensor of trust assertions
v_s, v_o	Trust subject and object $(v_s, v_o \in V)$
$\mathbf{A}^{K\|C}_{s,o}$	Trust assertion from v_s to v_o
$T^{\text{exp}}_{s,o}$	Experience-based trust value from v_s to v_o
$T^{\text{know}}_{s,o}$	Knowledge-based trust value from v_s to v_o
$\mathbf{k}^{\text{x}}_{s,o}, \mathbf{c}^{\text{x}}_{s,o}$	Evidence and context vector for $T^{\text{x}}_{s,o}$ (x $\in$ {exp, know})
$DT_{s,o}$	Direct trust value from v_s to v_o
$ST_{s,o\|p}$	Serial trust value from v_s to v_o along trust path p
v_r, V_c	Reference entity; set of common evaluators
$CT_{s,o\|r,V_c}$	Co-referential trust value based on v_r and V_c
$PT_{s,o\|\Omega}$	Parallel trust value over set of trust channels Ω

Here, K and C denote the trust evidence space and the trust context space, respectively. Specifically, the trust context space C comprises dimensions characterizing the operating conditions that may affect the fulfillment of the requested service, including the communication conditions, service requirements, and resource constraints. The trust evidence space K consists of multiple distinct evidence dimensions used to quantify the trustworthiness of trust entities. Each edge $e_{s,o} \in E$ is associated with a trust context vector $\mathbf{c}_{s,o}$, and a trust evidence vector $\mathbf{k}_{s,o} \in K$, referring to the specific values of $e_{s,o}$ across these dimensions.

For each direct trust relationship $e_{s,o}$, trust quantification maps the associated trust evidence and trust context to the corresponding trust assertion $\mathbf{A}^{K|C}_{s,o}$. According to the source of the underlying evidence, this assertion consists of two complementary components, namely experience-based trust $T^{\text{exp}}_{s,o}$ and knowledge-based trust $T^{\text{know}}_{s,o}$. The former reflects the trust object's behavioral performance observed by the trust subject through direct interactions, whereas the latter supplements such direct experience with verifiable information about the object, thereby providing an additional basis for assessing its capability and reliability when direct records are limited or uncertain. Levering the trust assertions in the trust graph, trust evaluation refers to the process of driving a trust value to measure the degree to which v_s can rely on v_o to fulfill the assigned service role, with the evaluation result denoted by Trust(v_s, v_o).

Furthermore, we define a trust path as a sequence of consecutive edges connecting a subject v_s to an object v_o, where $v_s, v_o \in V$ and $s \neq o$. It is denoted as a chain structure:

$$p : v_s \to \cdots \to v_{x_l} \to v_o, \tag{3}$$

where $l \geq 0$ is the number of the intermediate entities. When $l = 0$, i.e., $p : v_s \to v_o$, it indicates that v_s and v_o are directly connected, and the trust path exactly corresponds to the edge $e_{s,o}$ in the graph. When $l \geq 1$, the intermediate entities satisfy

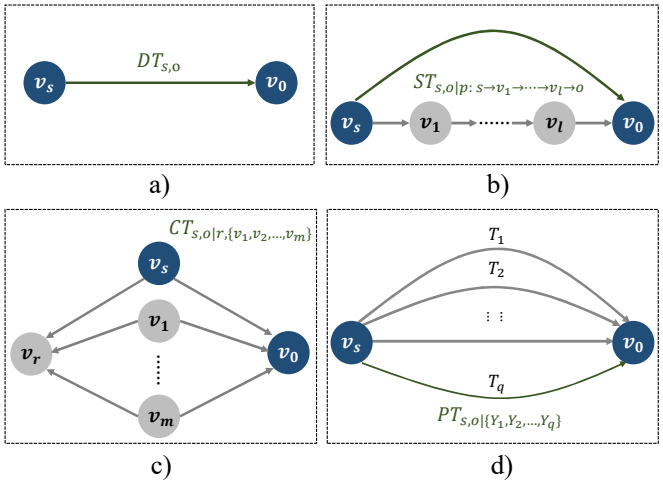


Fig. 2. Illustration of four TRUs in O-RAN trust graph: a) DTRU for direct trust, b) STRU for serial trust, c) CTRU for co-referential trust, and d) PTRU for parallel trust.

$x_i \in \{1, 2, \ldots, N\}$ and $x_i \neq s, o$, for $i = 1, \ldots, l$,and all x_i being distinct. Here, the symbol $\to$ denotes the trust flow between two directly connected entities in the trust graph. Additionally, we introduce a generalized form of trust path, namely trust channel between v_s and v_o, which is denoted as Y. It is a sub-graph that describes a trust association from v_s to v_o, encompassing all trust entities that directly participate in constructing this association, and the direct trust relationships between these entities. By this definition, a trust path is a specific type of trust channel.

Together, the sets V, E, and tensor $\mathbf{A}^{K|C}$, collectively form a structured trust graph considered in the O-RAN system. Meanwhile, the trust path p and the trust channel Y provide tools for analyzing the trust graph. We summarize the essential variables in Table I.

B. Trust Relationship Units

In O-RAN, the analysis of trust relationships arising from service provisioning is challenging due to the structural complexity of trust graphs [31]. These graphs encompass a multitude of entities with diverse service roles and multiple categories of trust relationships. Consequently, trust paths between entities often exhibit extensive branching and potential loops. To simplify, we introduce the concept of the trust relationship unit and define four fundamental TRUs, as shown in Fig. 2.

1) Direct TRU (DTRU): The DTRU characterizes the trust relationship established when v_s directly relies on a service, output, or function of v_o. Each edge in the trust graph therefore constitutes a DTRU. The strength of this relationship is measured by the direct trust value $DT_{s,o}$, which is derived from the corresponding trust assertion $\mathbf{A}^{K|C}_{s,o}$.

2) Serial TRU (STRU): The STRU characterizes an indirect trust relationship between v_s and v_o that relies on a trust path $p : v_s \to v_{x_1} \to \cdots \to v_{x_l} \to v_o$, where $l \geq 1$ denotes the number of intermediate entities. The strength of this trust relationship is quantified as the serial trust value, denoted by $ST_{s,o|p}$. Let $v_s = v_{x_0}$ and $v_o = v_{x_{l+1}}$, with $DT_{x_{i-1},x_i}, i \in \{1, \ldots, l+1\}$ denoting the direct trust between adjacent entities along the path. We define a serial operator, denoted as $\otimes$, to combine these direct trust values along the path, and formalized the serial trust value as

$$ST_{s,o|p} \triangleq \bigotimes_{i=1}^{l+1} DT_{x_{i-1},x_i}. \tag{4}$$

The STRU embodies the propagation of trust throughout the network. For example, an O-CU that cannot directly observe an O-RAN radio units (O-RUs) may evaluate the trustworthiness of the O-RU through the associated O-DU, which maintains direct service interactions with both entities.

3) Co-referential TRU (CTRU): The CTRU characterizes another type of indirect trust relationship through a trusted reference entity v_r and a set of shared evaluators $V_c = \{v_{x_1}, \ldots, v_{x_m}\}, m \geq 2$. The reference entity undertakes an identical or comparable service role to v_o and has established direct trust with v_s, with the trust value denoted by $DT_{s,r}$. Each shared evaluator has established trust relationship with both v_s and v_o. For the i-th shared evaluator, the direct trust values are respectively denoted as $DT_{x_i,r}$ and $DT_{x_i,o}$, $i \in \{1, \ldots, m\}$. These paired assessments reflect the similarity of service fulfillment performance of v_r and v_o. A co-referential operator, denoted by $\odot$, is defined to combine the existing $DT_{s,r}$ with the similarity reflected in these paired service assessments, yielding the co-referential trust value as

$$CT_{s,o|r,V_c} \triangleq DT_{s,r} \odot \{(DT_{x_i,r}, DT_{x_i,o})\}_{i=1}^{m}. \tag{5}$$

The CTRU provides an approach for trust establishment without relying on direct interaction history or explicit trust propagation paths. For example, before a UE relies on a newly connected O-DU from another vendor to provide radio access, it may select a previously interacted O-DU as the reference entity. Other UEs that have interacted with both O-DUs can provide paired assessments, thereby supporting the initial trust inference for v_o.

4) Parallel TRU (PTRU): The PTRU represents a trust relationship established through multiple indirect trust channels $\Omega = \{Y_1, Y_2, \ldots, Y_q\}, q \geq 2$. These channels may involve different intermediaries, trust evidence, or service context. Each channel is an STRU or CTRU, with the trust value derived from the i-th channel denoted by T_i. Hence, we have $T_i \in \{ST_{s,o|p}, CT_{s,o|r,V_c}\}, i \in \{1, \ldots, q\}$. We introduce a parallel operator, denoted by $\oplus$ to aggregate these trust values into an overall measure. The parallel trust value, denoted by $PT_{s,o|\Omega}$, is defined as

$$PT_{s,o|\Omega} \triangleq \bigoplus_{i=1}^{q} T_i. \tag{6}$$

For example, a UE may assess a candidate O-DU through multiple serial service dependency paths via different recommender UEs.

Through the definition of the above four TRUs, we clarify the establishment ways of different trust relationships, providing support for trust evaluation.

IV. Trust Quantification

A. Experience-based Trust

Experience-based trust $T_{s,o}^{\text{exp}}$ is derived from the object's behavioral performance in historical interactions. Accordingly, we define the experience-based trust evidence vector $\mathbf{k}_{s,o}^{\text{exp}}$ and the context vector $\mathbf{c}_{s,o}^{\text{exp}}$ as follows:

$$\mathbf{k}_{s,o}^{\text{exp}} \triangleq (\text{SR}_{s,o}, \text{RL}_{s,o}), \tag{7}$$

$$\mathbf{c}_{s,o}^{\text{exp}} \triangleq \left(p_{s,o}^{\text{c}}, d_{\max}\right). \tag{8}$$

where $\text{SR}_{s,o}$ denotes the service success ratio, which measures the functional fulfillment reliability of v_o after accounting for the environmental support probability $p_{s,o}^{\text{c}}$, and $\text{RL}_{s,o}$ denotes the response latency score, which measures the service timeliness relative to the service-specific delay requirement $d_{\max}$.

We refer to a service fulfillment event as an observable interaction in which v_o performs its assigned service role and produces an assessable outcome for v_s. An event is considered functionally successful if the resulting service, output, or function satisfies the predefined service requirements. Let n_s and n_f denote the number of successful and failed events, respectively. An observed service failure may result from either inadequate fulfillment by v_o or unfavorable service contexts. We therefore define $p_{s,o}^{\text{c}}$ as the representative probability that these operating conditions support successful service fulfillment, assuming that v_o correctly performs its assigned service role. This probability is determined according to the environmental factors relevant to the corresponding service provision under evaluation.

For example, for an event directly involving wireless data transmission, $p_{s,o}^{\text{c}}$ can be instantiated as the packet success probability under the representative channel conditions. Let $|H(f)|^2$ and $N(f)$ denote the channel power gain and the noise power spectral density at frequency f, respectively. Let B denote the effective channel bandwidth, and P_{tx} the transmission power committed by v_o. Then the signal-to-noise ratio (SNR), represented by γ, is calculated as $\gamma = \frac{|H(f)|^2 P_{\text{tx}}}{N(f)B}$. The bit error rate is derived from the SNR γ, expressed as $b = \xi(\gamma)$, where $\xi(\cdot)$ denotes the mapping function determined by the adopted coding strategy, modulation scheme and channel model. Denote the average packet length in the considered application scenario as n_l. Assuming independent bit errors, the environmental support probability is given by $p_{s,o}^{\text{c}} = (1-b)^{n_l}$.

Although $p_{s,o}^{\text{c}}$ reflects the degree of support provided by the current environmental conditions for successful service fulfillment, an intermediate probability may still indicate an ambiguous environmental state. We therefore further calculate its entropy, denoted by $u_{s,o}$,

$$u_{s,o} = -p_{s,o}^{\text{c}} \log_2 p_{s,o}^{\text{c}} - (1-p_{s,o}^{\text{c}}) \log_2 (1-p_{s,o}^{\text{c}}). \tag{9}$$

This entropy measure captures the ambiguity in attributing interaction outcomes to environmental factors versus behavioral factors. Let n_f^{c} represent the context-adjusted failure count attributable to the object's behavior rather than environmental causes. It is given by

$$n_f^{\text{c}} = n_f p_{s,o}^{\text{c}} \left(1 - \beta u_{s,o}\right), \tag{10}$$

where $\beta \in [0,1)$ controls how strongly environmental ambiguity reduces the attribution of failures to the trust object. (10) implies that, failures observed under strong and unambiguous environmental support (i.e., with high $p_{s,o}^{\text{c}}$ and low $u_{s,o}$) are more confidently attributed to the object, whereas failures occurring under weak or highly ambiguous environmental conditions are discounted. Successful events are retained without discounting, as they constitute positive evidence regardless of environmental conditions. Then, Bayesian smoothing is introduced to enhance robustness. The contextualized $\text{SR}_{s,o}$ is thus quantified as

$$\text{SR}_{s,o} = \frac{n_s + 1}{n_s + n_f^{\text{c}} + 2} \times \frac{n_s}{n_s + \delta}, \tag{11}$$

where the term $\frac{n_s}{n_s+\delta}$ is a penalty to adjust the trust value in limited experience scenarios, and $\delta > 0$ is a smoothing constant. This ensures that an object attempting to mask misbehavior behind environmental noise must still accumulate sufficient successful interactions to achieve a high SR value.

The response latency score $\text{RL}_{s,o}$ quantifies the timeliness of the service provided by v_o, serving as a complementary measure to functional success. Specifically, shorter delays result in higher scores, which in turn contribute to higher trust. The contextualized $\text{RL}_{s,o}$, is given by

$$\text{RL}_{s,o} = \max(0, \frac{d_{\max} - d_{s,o}}{d_{\max}}), \tag{12}$$

where $d_{s,o}$ refers to the average time delay between request initiation and response reception, $d_{\max}$ is the maximum allowable fulfillment delay specified for the corresponding service type.

Finally, the experience-based trust can be quantified as

$$T_{s,o}^{\text{exp}} = \lambda_{\text{exp}} \text{SR}_{s,o} + (1 - \lambda_{\text{exp}}) \text{RL}_{s,o}, \tag{13}$$

where $\lambda_{\text{exp}} \in (0,1)$ is a coefficient to balance the functional fulfillment reliability and service timeliness requirements.

B. Knowledge-based Trust

Knowledge-based trust $T_{s,o}^{\text{know}}$ is quantified through externally available information associated with the trust object. It complements the direct interaction history. To formalize this, we define the knowledge-based trust evidence vector $\mathbf{k}_{s,o}^{\text{know}}$ and the context vector $\mathbf{c}_{s,o}^{\text{know}}$ as follows:

$$\mathbf{k}_{s,o}^{\text{know}} \triangleq (\text{FR}_{s,o}, \text{RS}_{s,o}), \tag{14}$$

$$\mathbf{c}_{s,o}^{\text{know}} \triangleq \left(\kappa_{s,o}, \mathbf{r}_{s,o}^{\text{req}}\right). \tag{15}$$

where $\text{FR}_{s,o}$ represents the service fulfillment reputation, which reflects external feedback generated by other trust entities from their first-hand service fulfillment records with v_o. The confidence assigned to such feedback decays with its age, and the decay rate is controlled by the service criticality factor $\kappa_{s,o} \in [0,1]$. A larger $\kappa_{s,o}$ indicates that an incorrect or interrupted service contribution by v_o may cause more severe

consequences and therefore imposes stricter freshness requirements on the collected feedback. Moreover, $\mathrm{RS}_{s,o}$ represents the resource sufficiency score, which measures the margin of currently available resources at v_o beyond the minimum requirements $\mathbf{r}_{s,o}^{\text{req}}$ for the service under quantification. Notably, the service criticality, and resource requirements are obtained from the corresponding service profile maintained by the service management and orchestration (SMO).

Specifically, $\mathrm{FR}_{s,o}$ aggregates v_o's historical performance across the network. An object with a higher reputation is more likely to maintain reliable behavior, as past performance reflects both willingness and capability to fulfill obligations. Notably, rather than being a self-declared rating, feedback is derived from service fulfillment events according to (13). Let $T_{j,o}^{\text{exp}} \in (0,1], j \in \{1,...,m\}$ denote the latest experience-based trust value of v_o provided by v_j. Let $\tau_{j,o}$ denote the time elapsed since the feedback was generated or last updated. To account for contextual relevance, only feedback records associated with the same service category are considered. Let $L_{\max} > 0$ denote the longest historical period over which external feedback may be considered, as configured by the system. Accordingly, the set of admissible feedback records is $F_{s,o} = \{j \in \{1,...,m\} | \tau_{j,o} < L_{\max}\}$. In addition, a more critical service requires more recent external feedback. The freshness weight is then adjusted by its service criticality factor $\kappa_{s,o}$. Let $L_{s,o} = L_{\max}/(1+\kappa_{s,o})$. When $F_{s,o} \neq \emptyset$, the service fulfillment reputation is then calculated as

$$\mathrm{FR}_{s,o} = \frac{\sum_{j \in F_{s,o}} \left(\exp\left(-\tau_{j,o}/L_{s,o}\right) \times T_{j,o}^{\text{exp}}\right)}{\sum_{j \in F_{s,o}} \exp\left(-\tau_{j,o}/L_{s,o}\right)}. \tag{16}$$

When $F_{s,o} = \emptyset$, we set $\mathrm{FR}_{s,o} = 0.5$, corresponding to a neutral prior in the absence of external observations. Thus, more recently generated feedback contributes more significantly to $\mathrm{FR}_{s,o}$. A larger $\kappa_{s,o}$ imposes stricter freshness requirements on more critical services.

On the other hand, resource constraints are another key factor affecting their behavioral reliability. Trust entities with sufficient resources are more likely to fulfill commitments continuously and stably. Resource exhaustion may lead to service interruption or performance degradation, even when v_o has good intentions. Notably, the same resource level exhibits different degrees of sufficiency under different service requirements. Considering there are $x (x \in \mathbb{N}^+)$ critical resource types (e.g., computing capacity, storage, or transmission bandwidth). For the i-th type, let the object's current amount and the minimum requirement be denoted as $r_{o,i}^{\text{ava}}$ and $r_{s,o,i}^{\text{req}}$, respectively, $i \in \{1,...,x\}$, with $\mathbf{r}_{s,o}^{\text{req}} = (r_{s,o,1}^{\text{req}}, ..., r_{s,o,x}^{\text{req}})$. The minimum requirements of all critical resources are first enforced as service admission constraints, i.e., $r_{o,i}^{\text{ava}} \geq r_{s,o,i}^{\text{req}}, \forall i \in \{1,...,x\}$. Based on this prerequisite, the contextualized $\mathrm{RS}_{s,o}$ can be calculated as

$$\mathrm{RS}_{s,o} = \min_{i=1}^{x} \left(\min\left(1, \frac{r_{o,i}^{\text{ava}}}{\zeta r_{s,o,i}^{\text{req}}}\right)\right), \tag{17}$$

where $\zeta \geq 1$ represents the desired resource margin to support sustained and stable service provision. This formula indicates that the most scarce resource determines the overall capability.

Finally, we adopt the geometric mean to fuse the above two dimensions, ensuring that a deficiency in either dimension cannot be fully compensated by the other. The knowledge-based trust $T_{s,o}^{\text{know}}$ is then quantified as

$$T_{s,o}^{\text{know}} = (\mathrm{FR}_{s,o})^{\lambda_{\text{know}}} \cdot (\mathrm{RS}_{s,o})^{(1-\lambda_{\text{know}})}, \tag{18}$$

where $\lambda_{\text{know}} \in (0,1)$ is a balancing coefficient. (18) yields a high knowledge-based trust value only when the object exhibits both reliable service performance and sufficient resources, thereby avoiding overly optimistic assessments based on a single favorable dimension.

V. Service-Oriented Trust Evaluation in O-RAN

A. Evaluation for Directly Interacting Entities

When the trust subject v_s has direct interaction records with the trust object v_o, their relationship forms a DTRU, and the corresponding direct trust value serves as the trust evaluation result. As service interactions continue, the trust relationship may evolve with changes in the service behavior and operating conditions. Therefore, the direct trust value needs to be continuously maintained throughout the evaluation process, and updated by incorporating the latest available service-related evidence. To avoid overly frequent updates while preventing the trust value from becoming outdated, we trigger a direct trust update either when the cumulative number of newly observed service fulfillment events reaches a predefined threshold, or when the elapsed time since the previous update reaches the maximum waiting time. Formally, let t_0 be the initial time when the service interaction between v_s and v_o is established, and t_k the instant at which the k-th trust update is triggered, where $k \geq 1$. We denote the trust value after the k-th update by $DT_{s,o}^{(k)}$, and the initial direct trust value prior to the first update by $DT_{s,o}^{(0)}$.

For the k-th update, evidence is accumulated over the interval $(t_{k-1}, t_k]$, with duration $\triangle_k = t_k - t_{k-1}$. During this interval, the experience-based trust $T_{s,o}^{\text{exp}}$ is quantified using the newly observed service fulfillment events between v_s and v_o, whereas the knowledge-based trust $T_{s,o}^{\text{know}}$ is derived using the latest external reputation and resource state available at t_k. These two complementary components are first combined into an instantaneous trust estimate, denoted by $\widehat{DT}_{s,o}^{(k)}$, which reflects the current reliability of v_o from the perspective of v_s:

$$\widehat{DT}_{s,o}^{(k)} = w_{\text{e}} T_{s,o}^{\text{exp}} + (1 - w_{\text{e}}) T_{s,o}^{\text{know}}, \tag{19}$$

where $w_{\text{e}} \in (0,1)$ is a service-specific coefficient that specifies the relative importance assigned to first-hand service fulfillment experience. Then, the instantaneous trust estimate is further integrated with the previously established direct trust $DT_{s,o}^{(k-1)}$ to capture trust evolution across consecutive actual updates. The updated $DT_{s,o}^{(k)}$ is then expressed as

$$DT_{s,o}^{(k)} = e^{-\eta \triangle_k} DT_{s,o}^{(k-1)} + (1 - e^{-\eta \triangle_k}) \widehat{DT}_{s,o}^{(k)}, \tag{20}$$

where $\eta > 0$ is a time decay coefficient controlling the influence of historical trust. As the evidence accumulation interval $\triangle_k$ increases, the contribution of the previous trust state diminishes, and the trust update relies more heavily on

Algorithm 1 Trust subgraph construction algorithm

Require: trust subject v_s, trust object v_o, service context $\mathbf{c}$, depth limit $L_{\text{th}} \in \mathbb{N}^+$
Ensure: trust subgraph $G'_{s,o} = (V'_{s,o}, E'_{s,o}, (\mathbf{A}^{K|C})'_{s,o})$

Initialize $\mathcal{U}_{s,o}^{STRU} \leftarrow \emptyset$
Initialize $\mathcal{U}_{s,o}^{CTRU} \leftarrow \emptyset$
function TRUST_SEARCH(u, $depth$, $curpath$)
  if $u = v_o$ **then**
    Add $curpath$ into $\mathcal{U}_{s,o}^{STRU}$
    return
  end if
  if $depth \geq L_{\text{th}}$ **then**
    return
  end if
  for all $v \in \text{Neighbors}(u, \text{out})$ **do**
    if $v \notin curpath$ **and** (u, v) is relevant to $\mathbf{c}$ **then**
      $curpath.\text{push}(v)$
      TRUST_SEARCH(v, $depth + 1$, $curpath$)
      $curpath.\text{pop}()$
    end if
  end for
end function
TRUST_SEARCH(v_s, 0, $[v_s]$)
for all $v_r \in \text{Neighbors}(v_s, \text{out})$ **do**
  if v_r performs the comparable role as v_o **then**
    $V_c^{(r,o)} \leftarrow \{v_x | (v_x, v_r) \in E, (v_x, v_o) \in E\}$
    Add $(v_r, V_c^{(r,o)})$ into $\mathcal{U}_{s,o}^{CTRU}$
  end if
end for
$\mathcal{U}_{s,o} \leftarrow \mathcal{U}_{s,o}^{STRU} \cup \mathcal{U}_{s,o}^{CTRU}$
$G'_{s,o} \leftarrow \bigcup_{U \in \mathcal{U}_{s,o}} U$
return $G'_{s,o} = (V'_{s,o}, E'_{s,o}, (\mathbf{A}^{K|C})'_{s,o})$

the latest evidence. This allows the trust value to adapt to potential changes in service behavior. The resulting $DT_{s,o}^{(k)}$ represents the trust evaluation result after the k-th update.

B. Evaluation for Indirectly Interacting Entities

When the trust subject v_s lacks sufficient direct service records with the trust object v_o, it exploits direct trust relationships maintained by other entities to perform indirect trust evaluation. The evaluation follows a three-step process: trust subgraph construction, trust propagation and inference through valid TRUs, and aggregation of multiple indirect trust channels when necessary.

1) Trust Subgraph Construction: The first step is to identify the existing direct trust relationships relevant to the current evaluation, i.e., those that are compatible with the corresponding service context, and participate in forming valid trust channels from v_s to v_o, thereby constructing an evaluation-specific trust subgraph, denoted by $G'_{s,o}$.

Specifically, each valid STRU or CTRU connecting v_s and v_o constitutes an indirect trust channel, while multiple such channels can subsequently be combined into a PTRU. As detailed in Algorithm 1, for an STRU, v_s searches for directed paths consisting of consecutive direct trust relationships from v_s to v_o. Each distinct valid path constitutes a separate STRU. To limit the uncertainty and search overhead introduced by long-distance propagation, the path length is bounded by the maximum hop threshold, denoted by L_{th}. Moreover, each trust entity is allowed to appear at most once in a single path. If an entity recurs along a path (e.g., the path $p : v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow v_1$), the subsequent portion of the path will be discarded. This constraint eliminates cycles and avoids trust loops. For a CTRU, v_s first identifies eligible reference entities v_r that are directly trusted by and perform the same or a comparable service role as v_o. For each reference entity, v_s collects paired direct trust assertions from entities that have evaluated both v_r and v_o, and the resulting set constitutes a separate CTRU associated with v_r. Finally, all valid STRUs and CTRUs found, together with their constituent entities, direct trust relationships, and trust assertions, constitute the evaluation-specific trust subgraph $G'_{s,o}$.

2) Trust Propagation and Inference: With the graph simplified, indirect trust values are derived through the propagation and inference of direct trust values in the identified STRUs and/or CTRUs.

Firstly, for scenarios when v_s lacks direct service fulfillment records with a cross-vendor or cross-domain entity v_o, while intermediate O-RAN entities possess first-hand service records with adjacent entities, a trust path may be formed as, $p : v_s \rightarrow v_{x_1} \rightarrow ... \rightarrow v_{x_l} \rightarrow v_o, l \in \{1, ..., L_{\text{th}} - 1\}$. In this case, the direct trust value at the final hop can be propagated back to the trust subject through the cooperation of intermediate entities. As the resulting indirect trust is jointly constrained by all direct trust relationships along the path, any weak trust relationship limits the trust value that can ultimately be put on v_o. We therefore combine the constituent direct trust values multiplicatively. Additionally, the certainty of an indirectly derived trust result naturally decreases as more intermediate entities are involved. Thus, we introduce a path attenuation factor, denoted as $\alpha \in [0, 1]$. It is defined as the proportion of trust preserved after a single relay. The serial trust is then formalized as

$$ST_{s,o|p} = \prod_{i=0}^{l} \left(DT_{x_i, x_{i+1}}\right) \times \alpha^l, \tag{21}$$

where $v_{x_0} = v_s, v_{x_{l+1}} = v_o$, and α^l captures the cumulative attenuation along the path p.

Beyond path-based propagation, O-RAN may contain service entities from different vendors or administrative domains that perform the same or comparable service roles. When v_s lacks sufficient direct service fulfillment experience with the object v_o, it may infer trust in v_o by referring to a comparable service entity v_r that it already trusts, with the corresponding direct trust value denoted by $DT_{s,r}$. For this scenario, the co-referential trust can be inferred through a CTRU. As introduced in Section III-B, the shared evaluators are denoted as $V_c = \{v_{x_1}, \ldots, v_{x_m}\}, m \geq 2, m \in \mathbb{Z}$, where each entity has established trust relationship with both v_r and v_o. An evaluator whose assessments of the reference entity v_r is more consistent with those of v_s provides more valuable reference for v_s. Accordingly, we define the reference weight

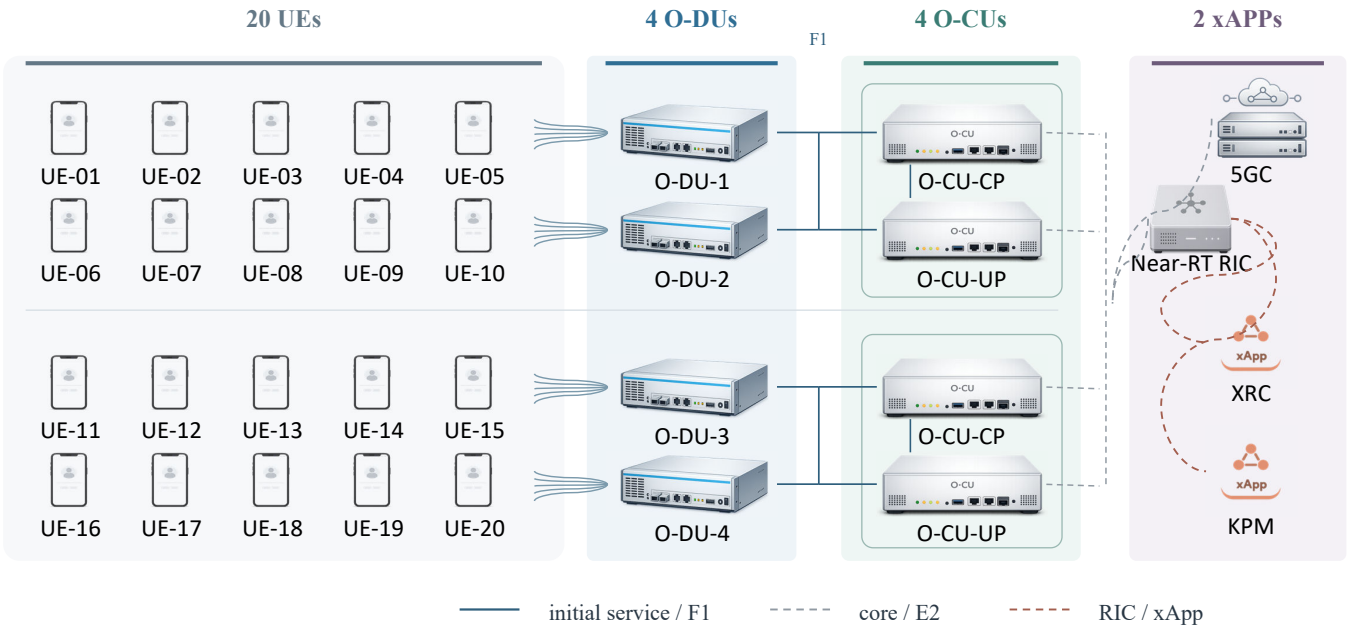


Fig. 3. Topology of the OAI-based O-RAN prototype, consisting of 20 UEs, 4 O-DUs, 2 O-CU-CPs, 2 O-CU-UPs, and 2 monitoring xApps.

of the i-th evaluator as $y_i^{(r)} = 1 - |DT_{s,r} - DT_{x_i,r}|$. Based on this weight, we define the similarity between the reference v_r and object v_o as $\text{Sim}_{r,o|V_c}$, given by

$$\text{Sim}_{r,o|V_c} = \frac{1}{m}\sum_{i=1}^{m} y_i^{(r)} \left(1 - |DT_{x_i,r} - DT_{x_i,o}|\right), \tag{22}$$

where the term $1-|DT_{x_i,r} - DT_{x_i,o}|$ captures the consistency between the evaluator's trust values towards v_r and v_o. The reference weight ensures that evaluators whose trust preferences are more compatible with v_s contribute more to the final similarity measure. Thus, the co-referential trust $CT_{s,o|r,V_c}$ is computed as

$$CT_{s,o|r,V_c} = DT_{s,r} \times \text{Sim}_{r,o|V_c}, \tag{23}$$

which indicates that v_s transfers its existing trust in the reference entity v_r to the target entity v_o, scaled by the degree of similarity between the two entities, as perceived by evaluators whose judgment criteria are consistent with v_s. Since $0 \leq \text{Sim}_{r,o|V_c} \leq 1$, it follows that $CT_{s,o|r,V_c} \leq DT_{s,r}$, making co-referential trust inference inherently conservative in O-RAN system.

3) Trust Aggregation: The final step is to aggregate the indirect trust values derived from the previous step into a unified evaluation result for v_o. In particular, when only one indirect trust value is available, it is directly retained as the final evaluation result without further aggregation. Otherwise, the trust channels associated with these indirect trust values collectively constitute a PTRU, These trust values are aggregated to obtain a parallel trust value $PT_{s,o}$, which serves as the trust evaluation result for v_o.

Let $\Omega_{s,o}$ denote the set of valid trust channels between v_s and v_o. The trust value derived from i-th channel is denoted by T_i, where $T_i \in \{ST_{s,o|p}, CT_{s,o|r,V_c}\}, i \in \{1, ..., q\}, q \geq 2$. Each channel represents an opinion regarding the trust state of v_o. Let $H \in \{0, 1\}$ be a binary random variable, where $H = 1$ indicates that v_o is trustworthy and $H = 0$ indicates that v_o is untrustworthy. Within each trust channel, T_i can be interpreted as the probability supporting the hypothesis $H = 1$, i.e., $p_i(H = 1) = T_i,\ p_i(H = 0) = 1 - T_i$. To mitigate the influence of trust channels with significant disagreements, we adopt a median-based weighting scheme. Specifically, let M_Ω denote the median of these trust values. The absolute deviation of T_i from the median M_Ω is thus written as $|T_i - M_\Omega|$. A smaller deviation represents that the corresponding trust result is more consistent with the majority. Then, the normalized aggregation weight of T_i, denoted by g_i, is defined as

$$g_i = \frac{1 - |T_i - M_\Omega|}{\sum_{j=1}^{q}\left(1 - |T_j - M_\Omega|\right)}. \tag{24}$$

The trust values from different channels are then aggregated through a weighted logarithmic opinion pool. Using the logarithmic pooling rule, the parallel trust $PT_{s,o|\Omega}$ is expressed as

$$PT_{s,o|\Omega} = \frac{\prod_{i=1}^{q}(T_i)^{g_i}}{\prod_{i=1}^{q}(T_i)^{g_i} + \prod_{i=1}^{q}(1 - T_i)^{g_i}}. \tag{25}$$

Equivalently, this rule combines the trust opinions by taking their weighted average in the log-odds domain. The contribution of each trust channel is adjusted according to its deviation from the median. Since the median is less sensitive to extreme values, an abnormal assessment cannot significantly shift the median reference, thereby mitigating the impact of a badmouthing attack [32], in which some malicious entities attempt to lower the aggregated trust value by providing abnormally low scores through a limited number of channels. For trust subjects and objects that are not directly connected, once the parallel trust calculation is completed, the final trust value is obtained.

VI. Utilizing Evaluated Trust in O-RAN

This value then serves as the basis for establishing or maintaining the corresponding service relationship in O-RAN service integration.

VII. Experimental Results

A. OAI-based Prototype

We implement a containerized O-RAN prototype based on OpenAirInterface (OAI) to evaluate the proposed trust framework. As shown in Fig. 3, our prototype consists of 30 components, including 20 UEs, 4 O-DUs, and other supporting components for service delivery and telemetry collection, such as O-CU-CPs, O-CU-UPs, the RAN intelligent controller (RIC), and xApps. In this prototype, each active UE-to-O-DU association gives rise to a service-oriented trust relationship, where the UE acts as the trust subject and the serving O-DU acts as the trust object. Initially, UEs are evenly attached to the four O-DUs. During operation, UEs continuously collect service-related evidence and evaluate the trustworthiness of the serving O-DUs. Each control round lasts 20 seconds. For every active UE-to-O-DU relationship, the prototype collects 150-packet service observations. When the trust value of the serving O-DU falls below a predefined threshold, the UE releases the current connection and establishes a new connection with another candidate O-DU based on the trust evaluation results. A selection module uses the resulting trust values to determine the serving O-DU for the next round. When the selected candidate O-DU differs from the current serving O-DU and the operational constraints allow the change, a controlled reattachment is performed and takes effect in the subsequent round.

Using this prototype, we validate the proposed O-RAN trust framework through experiments. We consider two types of service degradation scenarios. The first involves environment-induced shared service degradation, where multiple O-DUs experience simultaneous performance degradation while their service behaviors remain normal. The second considers service degradation caused by malicious service behaviors, where the O-DU-2 is configured to exhibit unreliable service behaviors, such as packet dropping. For comparison, three representative trust evaluation methods are selected: 1) BLTM [33]: evaluates trust by integrating multiple direct trust factors and recommendation trust, while considering link quality information during trust computation. 2) TEM-MENs [21]: evaluates trust between entities based on direct interactions when available; otherwise, it infers trustworthiness through trust propagation along one or multiple trust paths. 3) TaaS [24]: employs a reputation-based trust framework that calculates trust scores based on historical interactions, recommendations, feedback credibility, and SLA-related violation information. All methods use the same topology, initial attachment distribution, packet observation schedule, predefined service behavior vectors, and common operational constraints during the experiments to ensure a fair comparison. The experimental parameters are summarized in Table II.

Table II. Experimental settings and key parameters.

Parameters	Values
OAI nodes	30
UE / O-DU / O-CU-CP / O-CU-UP / xApp	20/4/2/2/2
Experiment rounds	30
Interactions per round	150
Maximum service delay $d_{\max}$	200 ms
Initial trust T_0	0.5
Trust threshold T_{th}	0.5
Experience-based trust $(\lambda_1, \beta, \delta)$	$(0.8, 0.5, 5)$
Knowledge-based trust (λ_2, ζ)	$(0.7, 1.1)$
DTRU (w_{e}, η)	$(0.7, 1.5)$

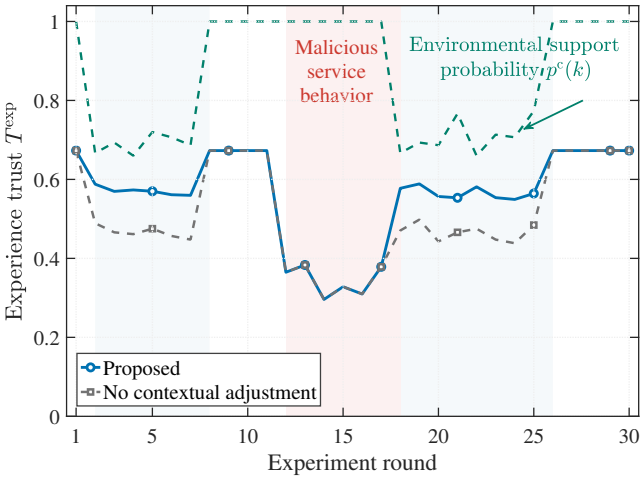


Fig. 4. Impact of contextual adjustment on experience-based trust quantification under varying service contexts.

B. Performance Analysis

We first evaluate the proposed experience-based trust quantification method with contextual adjustment. Fig. 4 presents the evolution of experience-based trust from a UE to an O-DU under fluctuating service contexts characterized by the environmental support probability $p^{\text{c}}(k)$. Environmental disturbances are introduced in rounds 2 to 7 and 16 to 21, while malicious packet dropping by the evaluated O-DU is introduced in rounds 10 to 15. We compare the proposed method with a baseline without contextual adjustment. During the initial rounds, the service context remains stable with a high $p^{\text{c}}(k)$, and both methods obtain similar trust values under normal service contexts. In rounds 2 to 7 and 16 to 21, $p^{\text{c}}(k)$ decreases, indicating degraded service contexts caused by the environment. As we can see, the proposed context-aware method only slightly decreases the trust value, whereas the method without contextual adjustment applies stronger penalties and causes trust values to fall below the threshold in some rounds. Then, during rounds 10 to 15, both methods show a rapid decrease in trust values as failures caused by malicious packet dropping accumulate. These results demonstrate that the proposed context-aware quantification method can reduce unnecessary trust penalties caused by contextual variations while remaining sensitive to service failures caused

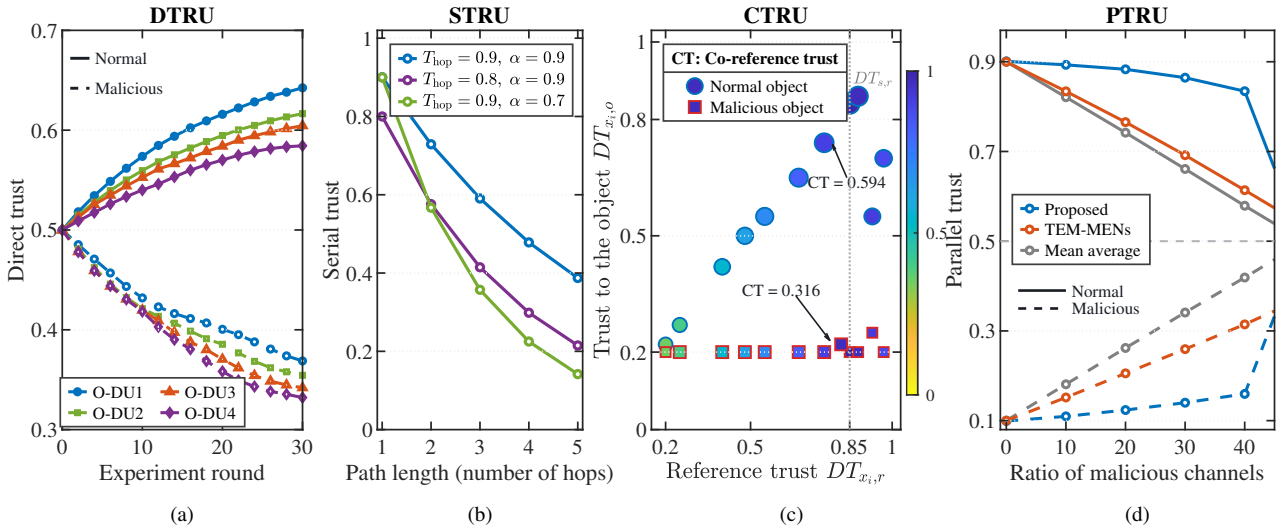


Fig. 5. Evaluation of different TRUs for characterizing diverse trust relationships. (a) Direct trust evolution for normal and malicious O-DUs in DTRUs. (b) Serial trust decay under varying T_{hop} and α in STRUs. (c) Co-referential trust inference for normal and malicious O-DUs in CTRUs. (d) Parallel trust under varying malicious channel ratios in PTRUs.

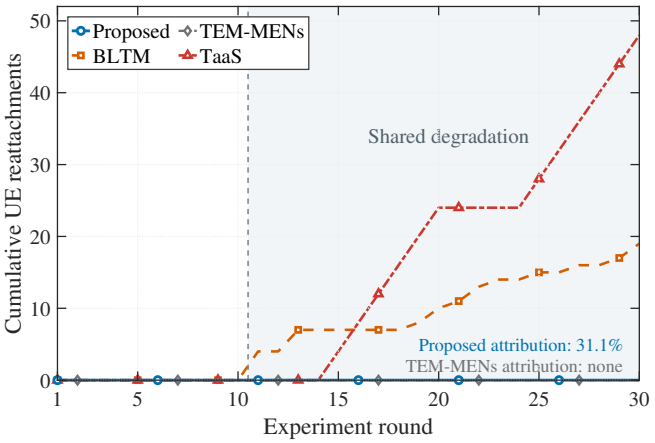


Fig. 6. Performance of UE reattachment decisions under shared service degradation.

by unreliable entities.

We further evaluate different TRUs for characterizing diverse trust relationships, as summarized in Fig. 5. First, Fig. 5(a) presents the evolution of direct trust in DTRUs during continuous interaction rounds for four O-DUs with normal and malicious service behaviors, respectively. The results show that the direct trust of normal O-DUs gradually increases with accumulated successful interactions, whereas that of malicious O-DUs decreases. This validates that the proposed DTRU can effectively reflect service fulfillment reliability and thus distinguish unreliable entities. Fig. 5(b) shows the serial trust across three STRUs with different hop direct trust value T_{hop} and attenuation factor α. Results demonstrate that the serial trust decays with the number of hops, and the decay becomes more severe as T_{hop} or α decreases. This confirms that any weakness in the path is progressively amplified along the chain and eventually constrains the final trust value. Fig. 5(c) illustrates the co-referential trust evaluation for a target O-DU under normal and malicious service behaviors. We select one O-DU as the reference entity and 12 UEs as common evaluators. The vertical dashed line indicates the subject's trust toward the reference entity $DT_{s,r}$. Each marker represents a direct trust pair $(DT_{x_i,r}, DT_{x_i,o})$ collected from a common evaluator, with its color representing the contribution weight of the corresponding evaluator in (22). It can be observed that evaluators whose assessments are closer to the dash line receive larger weights, as they may own more consistent evaluation criteria with the subject. Moreover, degraded service reduces the similarity between the reference and target entities, leading to a decrease in the co-referential trust from 0.594 to 0.316, both remaining below $DT_{s,r}$. This result demonstrates the effectiveness of our CTRU in performing conservative trust inference by leveraging the reference entity and common evaluators. Fig. 5(d) compares the parallel trust values obtained by different aggregation methods as the fraction of malicious trust channels increases. The results show that the proposed PTRU can maintain more stable trust evaluation, whereas TEM-MENs and the mean aggregation baseline suffer from larger deviations due to biased evaluations. This is because the proposed PTRU reduces the impact of inconsistent trust channels by adjusting their aggregation weights according to their deviations from the majority opinion.

We further investigate whether the proposed trust framework can support reliable service decisions in O-RAN under contextual variations and malicious behavior. At the end of each round, the UE could decide whether to switch to another O-DU and which one to choose, based on the trust evaluation results. Fig. 6 shows the cumulative number of UE reattachments to alternative O-DUs over 20 UEs under shared service degradation. As we can see, during the period of shared degradation

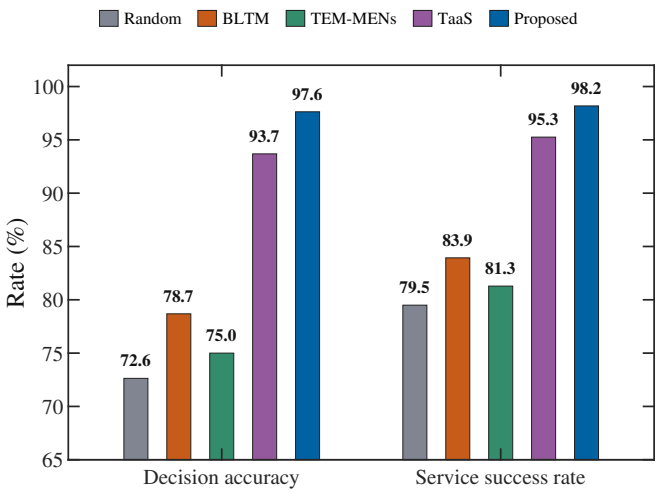


Fig. 7. Performance of UE reattachment decisions under malicious service behavior.

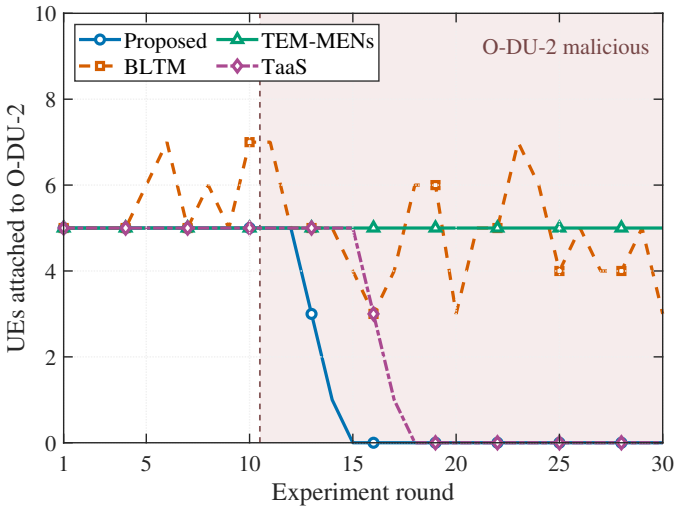


Fig. 8. Number of UEs attached to malicious O-DU-2.

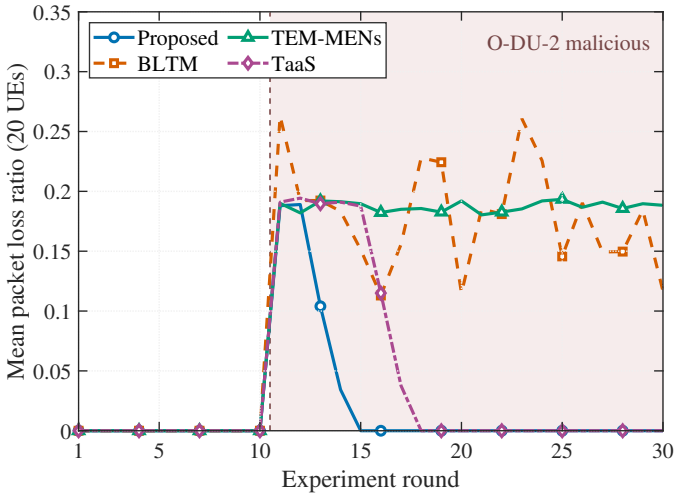


Fig. 9. Average packet loss ratio under malicious service behavior.

after round 10, the proposed method and TEM-MENs maintain zero reattachments, but for different reasons. The proposed method attributes part of the observed failures to contextual conditions, whereas TEM-MENs maintains relatively high trust values due to its resource-related trust metrics, but lacks explicit contextual attribution for service degradation. BLTM also shows limited adaptability to changing service contexts. In contrast, TaaS triggers the most frequent reattachments because SLA violations are directly reflected in service decisions without considering contextual factors.

Next, we examine the performance under malicious service behaviors. In our prototype, O-DU-2 is configured to exhibit malicious service behavior from round 11 onward. In each subsequent round, each UE updates its trust evidence based on the observed service behavior, computes trust values using the corresponding evaluation method, and selects one of the four O-DUs as its serving O-DU for the next round. A random selection strategy is also introduced as a baseline. The evaluation window contains 19 rounds after the malicious behavior is introduced, where 20 UEs perform O-DU selection in each round, resulting in 380 selection decisions for each method. We adopt the following two metrics to evaluate the performance: 1) Decision accuracy: the proportion of UE decisions that select an O-DU providing honest service. 2) Service success rate: the proportion of service interactions that are successfully completed after the selected O-DUs are applied. Fig. 7 compares the proposed framework, BLTM, TEM-MENs, TaaS, and random selection. The results show that the proposed framework achieves a decision accuracy of 97.63%, which outperforms BLTM and TEM-MENs by 18.90% and 22.60%, respectively. Compared with TaaS, the proposed framework achieves a service success rate of 98.2%, which corresponds to a 60% reduction in the derived service failure rate from 4.74% to 1.82%. This suggests that the proposed method improves the reliability of service selection among candidate entities by leveraging trust evaluation results.

Finally, Fig. 8 presents the number of UEs attached to malicious O-DU-2, and Fig. 9 shows the average packet loss rate (PLR) across all 20 UEs. As we can see, during the first 10 rounds of normal service, the PLR of all four trust methods remained nearly zero, indicating a stable service. After round 11, when O-DU-2 begins its malicious service, the PLR values of all methods increase, and the average PLR across the 20 UEs reaches about 19%. For the proposed framework, the trust evaluation result of the malicious O-DU gradually decreases, causing UEs initially served by O-DU-2 to switch to other O-DUs from round 13, and complete this switching process by round 15, after which the PLR decreases accordingly. TaaS also responds to the service degradation and completes the switching process by round 18, resulting in a slower recovery compared with the proposed framework. In contrast, BLTM exhibits unstable performance. As shown in Fig. 9, it sometimes switches back to the malicious serving O-DU-2. These results demonstrate that the proposed framework can select more trustworthy candidate O-DUs for service provision and thus improve the overall performance of O-RAN service delivery.

VIII. Conclusion

In this paper, we proposed a trust framework that models, quantifies, and evaluates trust relationships among O-RAN entities in participating service collaboration. We modeled service-oriented trust relationships through a trust graph and characterized different relationship patterns using TRUs. We quantified trust based on direct experience and verifiable knowledge with contextual adjustment, enabling trust values to capture the reliability of service fulfillment under different service conditions. We further performed trust evaluation for diverse trust relationships through TRU-based trust inference and aggregation, thereby supporting trustworthy collaborative services in O-RAN. Finally, we validated the proposed framework through experiments on a lab-built prototype. Results demonstrate its effectiveness in distinguishing unreliable service entities, informing service decisions, and improving O-RAN service reliability. The proposed framework provides a basis for reliable service provision in O-RAN.

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